

Vinay Shanker

 Portfolio |  Gmail |  LinkedIn |  LeetCode |  GitHub

PROFESSIONAL SUMMARY

Patent co-inventor and Software Engineer with experience in backend systems, distributed architecture, and full stack development. Co-invented Solo Tutor (patent filed at Jain University) and built production-grade systems using Spring Boot and React, including event-driven microservices, API gateway design, and RAG pipelines. Seeking full-time Software Engineer roles to build scalable backend systems and impactful products.

EDUCATION

Jain (Deemed-to-be University), Faculty of Engineering & Technology <i>B.Tech in Information Science & Engineering (Expected 2027) CGPA: 8.3</i>	Aug 2023 – Present Bengaluru, Karnataka
Bharatiya Vidya Bhavan's <i>Higher Secondary Education (12th Grade) 83%</i>	2021 – 2023 Bhimavaram, Andhra Pradesh

PROJECTS

Solo Tutor — AI-Powered Learning Platform | *Python, FastAPI, React, Supabase, pgvector* 2024 – Present

- Patent Filing In Progress.** Named co-inventor among five student inventors for an AI-powered learning platform at Jain University.
- Built a multimodal RAG pipeline for hallucination-free Q&A over documents and YouTube URLs using pgvector in Supabase, supporting retrieval across 1,000+ embedded document chunks.
- Engineered automated quiz generation from uploaded content and YouTube transcripts with Whisper-based audio fallback; system supports concurrent ingestion with sub-second query response.

Distributed Rate Limiter & API Gateway | *Java, Spring WebFlux, Redis, Docker*

- Built a high-throughput API Gateway using Token Bucket rate limiting with Spring WebFlux; load-tested to handle 10,000+ concurrent requests without thread exhaustion.
- Implemented atomic distributed state using Redis Lua scripts, ensuring race-condition-free updates with sub-10ms response times under sustained load.

Event-Driven E-Commerce Order Engine | *Spring Boot, Kafka, PostgreSQL, Docker*

- Designed fault-tolerant microservices (Order, Inventory, Payment) using the Saga Pattern to maintain data consistency across independent services.
- Orchestrated asynchronous event streams with Apache Kafka and integrated Zipkin for end-to-end distributed tracing across service boundaries.

TECHNICAL SKILLS

Languages: Java, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Spring Boot, Spring WebFlux, Spring MVC, Spring Security, Spring Data JPA, Hibernate, React.js

Authentication: JWT Auth

Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka

AI & Vector Search: Spring AI, RAG, pgvector

Developer Tools: Git, Docker, Maven, Gradle, JMeter, Zipkin, VS Code

ACHIEVEMENTS & EXTRACURRICULAR

- Co-inventor on Solo Tutor patent application (Jain University), recognised for contributions to AI-powered quiz generation and RAG pipeline modules.
- Self-directed study of distributed systems, independently implementing Token Bucket rate limiting, Saga Pattern, and Kafka event streaming beyond coursework.
- Actively practising DSA on LeetCode to strengthen problem-solving fundamentals for technical interviews.